# Quantitative Action Quality Assessment

Parting the Wild Horse's Mane: form-level sequences

## Segmented SMPL Sequences

One reference; a student cohort

**Reference: QxVvRcRn2TA**



Start ... End

**Student: 10**



Start ... End

Real tracking meshes
Canonical front view for display

## (a) Uniform-Anchor Pose Comparison

### Uniform Sampling

Reference anchors at 1 FPS

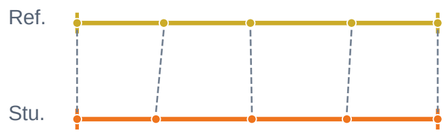


Include first and last frames

**K fixed reference anchors**

### SMPL-DTW Matching

Local cost: mean SO(3) distance




For each anchor: minimum local-cost student candidate on the DTW path

### Matched-anchor joint distances




23 local joints

**Mean Pose Distance**

Average over K anchors and 23 local joints; global orientation excluded.

## (b) Pelvis Motion Comparison

### Body-Relative Signals

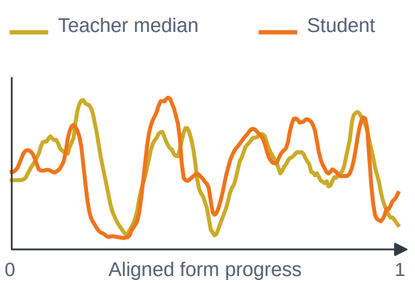
Pelvis relative to ankle midpoint

Height, lateral and forward motion

Normalize by body size

DTW-align to form progress

### Lateral pelvis trajectory




### Pelvis Deviation

Horizontal trajectory RMSE
Horizontal endpoint offset
Vertical trajectory RMSE

Teacher-calibrated excess

**Fixed Weighted Sum**

Pelvis reference: median of 10 teachers; robust leave-one-teacher-out calibration.

## Equal-Weight Borda Fusion

Rank students within the same form

**Pose rank**

Smaller mean distance is better

**Pelvis rank**

Smaller deviation is better

**50% pose + 50% pelvis**

$$Q = 0.5q_{\text{pose}} + 0.5q_{\text{pelvis}}$$

$$q = (N - r)/(N - 1)$$

## Relative Quality Ranking

Order by fused rank quality Q

Average ranks for ties; N > 1

Uniform sampling selects anchors only; DTW operates on the full sampled pose sequences.

Pose examples and curves use existing results. No absolute quality score or concrete final ranks are shown.